

Wrist Injuries

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Isabelle Schöffl in „North Star“, Frankenjura, Germany, Photo Kilian Reil

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7.1 Overview (Statistics)

Depending on the classification of injuries and overuse damage to the wrist region, complaints in this area account for approximately 7–10% of all climbing-related injury patterns [1–3]. This is the fourth most common injury region after finger, shoulder, lower extremity, and foot injuries and must therefore be considered when treating climbers and boulderers [1–3]. According to current figures, the most common symptom is the ulnar-sided overuse reaction of the wrist [3]. In addition, Dupuytren's contracture of the palm accounts for a

large proportion of chronic complaints of the metacarpus and can often be diagnosed, especially in long-time (older) athletes [4] (see also Chap. 15, “Climbing in Older Athletes.”).

7.2 Diagnosis

The initial diagnostic procedure for complaints of the wrist, metacarpus, and distal forearm area follows the usual criteria and should be structured as follows:

1. *Patient history*: How long have the complaints existed? After which strain did they first appear? Can the complaints be triggered by certain grip/pull shapes? Do the complaints radiate? Have the pain characteristics changed? Which treatments have been administered so far?
2. *Clinical examination*: Inspectorial abnormalities (swelling, color changes of the skin, etc.). Palpation of the corresponding region (pressure tenderness, etc.). Stress tests depending on the presumed diagnosis (hamate stress test, forced radial/ulnar duction, etc.). This should always be accompanied by a comparative examination of the opposite side!
3. *Imaging procedures*: Sonographic examination of the corresponding region, X-ray, CT, MRI diagnostics, depending on the suspected diagnosis.

Detailed anamnesis and a proper clinical examination are of utmost importance in the clarification of wrist pain and pathologies in the metacarpus and distal forearm. With appropriate knowledge of the various common clinical patterns and the respective stress tests, correct diagnoses can often be made with a relatively high degree of certainty. Unnecessary imaging procedures can thus be avoided. Regarding the clarification of pain in the area of the metacarpals, also refer to Chap. 9, “Finger Injuries.”

7.3 Bone Edema

Bone marrow edema (BME) of the carpus, the metacarpus, and the distal forearm is usually the result of chronic overloading, usually triggered

by repetitive, unphysiological movements [5–9]. Pathophysiologically, bone marrow edema corresponds to an increased accumulation of tissue fluid and blood within the respective bone as well as increased blood flow in the affected area [10]. It should be noted that the climbing-related stress-induced bone marrow edema is based on a different pathogenesis than transient bone marrow edema, ischemic osteonecrosis, vascular necrosis, or bone contusions [11–14]. The diagnosis of corresponding overuse damage has become more frequent in recent years due to the increasing use of MRI diagnostics [5, 6, 10].

Bone marrow edema can be divided into the following categories: mechanical bone edema (chronic overload, fractures, malpositions, degenerative), ischemic bone edema (osteonecrosis), inflammatory bone edema (rheumatoid arthritis, infectious), idiopathic (transient) bone edema, or bone edema due to malignancies. The most frequently affected areas for stress-related BME are usually static areas of the lower extremities. However, BME of the upper extremities, especially in the hand, are very rarely described in literature [8, 14–18].

In climbing, however, bone marrow edema is more frequent due to the continuous tensile load and the resulting shear forces in the hand, wrist, and forearm [5, 7]. Based on the complex anatomy of the carpus, overload reactions can affect all carpal bones as well as in the metacarpals and forearms. However, BME are frequently described in the following regions: lunate, capitate, hamate, bases of the metacarpals, distal radius, and distal ulna [5–7]. It should also be noted that the osseous reactions described differ from osteochondroses such as Kienböck’s disease and others. The differentiation to these disease patterns can sometimes be difficult (Fig. 7.1).

Affected climbing athletes commonly complain of a slow-onset and increasing pain around the affected bone, which usually improves to a certain extent when the load is reduced (rest). During examination, the pain can usually be triggered or intensified by palpation of the affected region. Specific stress tests (see hamate fractures, ulnocarpal pain reactions, and Chap. 9, “Finger Injuries”) can also be helpful [5]. Bone marrow edema generally remains inconspicuous in images produced by sonography, X-ray, and CT scans [10, 14, 19].

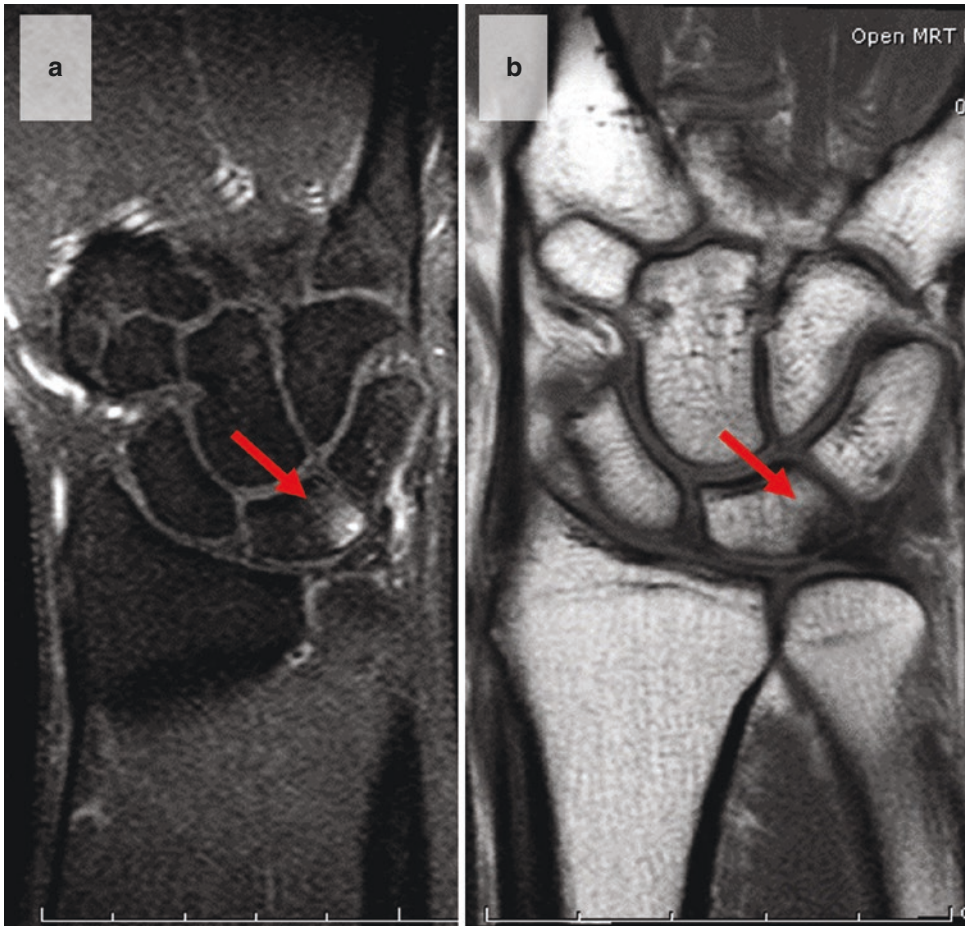


Fig. 7.1 MRI of bone marrow edema within the lunate (**a**, T2 weighting; **b**, T1 weighting)

However, MRI depicts the stress reaction within the affected bone. The therapeutic approach consists of immediate and long-term stress reduction with subsequent adjustment of training. Supplementary shockwave therapy can also help to promote healing; however, there is no corresponding evidence of this in the literature to date (Fig. 7.2).

After the symptoms have subsided, supporting circular wrist tape can be applied; however, this should only be used if it gives the athlete a feeling of stabilization [20].

It can be assumed that the bone marrow edema of the wrist will become increasingly important due to the increasing popularity of bouldering worldwide [21]. An early and correct diagnosis and sufficient treatment can prevent serious secondary conditions, such as stress fractures caused by overstrain [5–7, 19].

- **Etiology:** Bone marrow edema within the wrist is based on chronic overuse.
- **Onset:** Usually slow-onset and load-dependent complaints.
- **Clinic:** Circumscribed pain symptoms around the affected bone.
- **Diagnosis:** Clinical and magnetic resonance imaging (MRI).
- **Therapy:** Immediate stress reduction and long-term training change. Mostly long healing process.
- **Result:** Mostly complete healing possible.

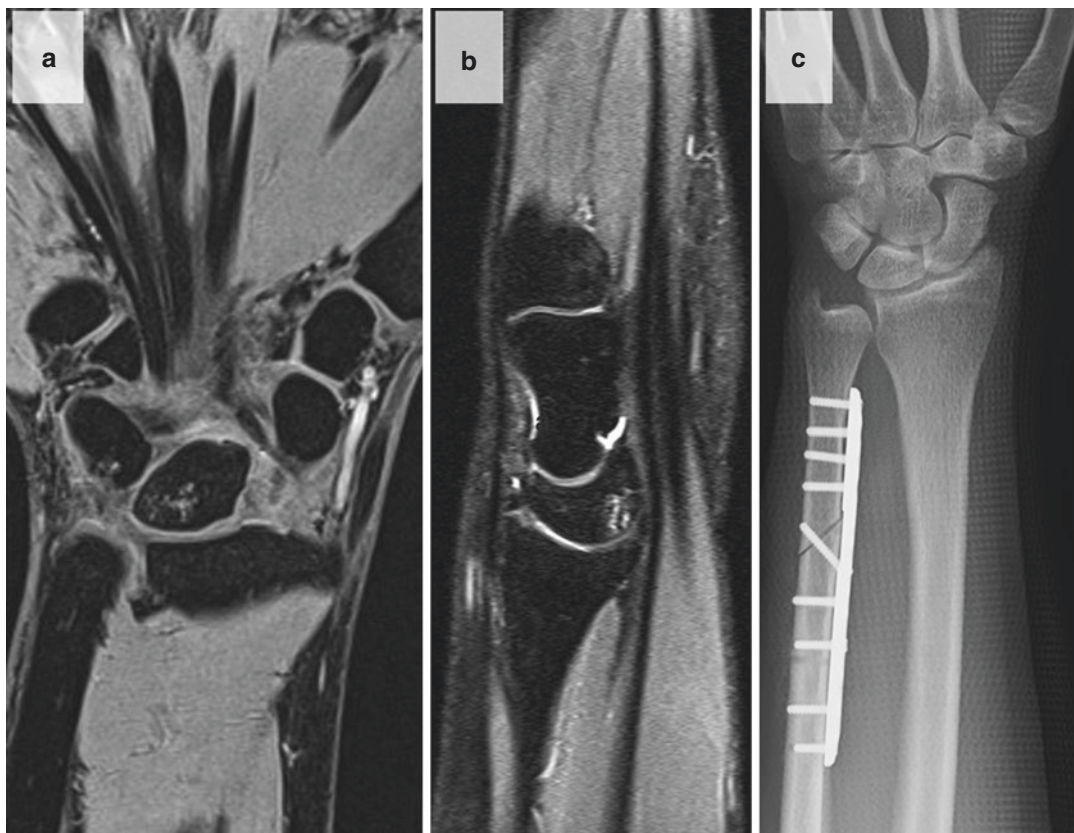


Fig. 7.2 MRI (a, b) and X-ray (c, left wrist) of a 30-year-old female bouldering athlete who had reported long-lasting and consistent pain even after having received arthroscopic debridement of the TFCC 2 years earlier (positive ulna variance). An extended shockwave therapy

of the wrist (lunate) was performed in addition to long rest. However, follow-up MRI and CT scans did not show sufficient bony healing within the lunate. Finally, an ulna shortening procedure was performed (c)

7.4 Hamate Fractures

Sport-related fractures of the hamate are usually caused by direct impact trauma (e.g., direct impact from handlebars (e.g., mountain biking) or from a club (e.g., baseball or golf)) [22–28]. These direct impact traumas are rare in climbing [5, 19], in which the cause is usually indirect overloading of the hamate [5, 22]. In particular, the hamate hook, the bony deflection pulley of the flexor tendons of the ring and little finger, is subjected to enormous stress due to the high contact pressure [5]. This high contact pressure

increases in ulnar-abducted hand positions (e.g., under grips) and under high tensile forces on the ring and little finger. Hamate fractures in climbing almost exclusively affect the hook and are divided depending on their location into type I(I) fractures (avulsion fractures at the tip of the hook), type I(II) fractures (center part of the hook), and type I(III) fractures (base of the hook) (classification according to Milk and Xiong) (Fig. 7.3) [5, 22].

The patient's medical history generally reports a slow-onset, load-dependent, and progressive pain on the volar and ulnar side of the

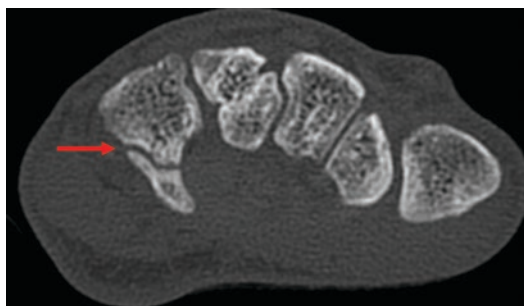


Fig. 7.3 CT scan of a hamate hook base fracture (see arrowhead)

wrist. It can also be helpful to administer a so-called hamate stress test, in which the patient is asked to flex his ring and little finger strongly against resistance (e.g., the examiner's finger) with an ulnar-abducted wrist (see Fig. 5.1. Fig. 7) [29]. An overload reaction or a fracture in the area of the hamate is indicated by significant pain in the affected area, which is less pronounced in a subsequent retest in the radially abducted wrist under the same load on the ring and little finger [29]. To confirm the diagnosis, immediate magnetic resonance imaging or computed tomography should be performed; conventional radiological examination can sometimes be misleading [5, 19, 30, 31].

The authors recommend that a conservative therapy attempt (initial consistent immobilization!) should always be carried out first [5, 22]. Since the hamate hook plays an essential role as a redirection of the flexor tendons (Dig. IV/V), immediate resection of the bone portion leads to a loss of strength; this should be avoided in climbers [32]. Conservative therapy often promises sufficient healing of the fracture if the patient is not participating in sports. In the case of insufficient bone healing, either spot-drilling of the fracture site (Fig. 7.4) or compression osteosynthesis using a Herbert screw should be considered (Fig. 7.5). A resection of the hamulus should only be performed in exceptional cases as it may lead to strength deficit (Fig. 7.6) [22, 30–33]. Regardless of the form of therapy, the training must be adjusted following the injury to avoid refractures or other injuries. A circular wrist tape can be considered after resuming sport [20].



Fig. 7.4 Spot-drilling of nonunion of the hamate using a Kirschner wire

- **Etiology:** Chronic overload reaction due to contact pressure of the flexor tendons. Initially bone edema in the hamate (hook); continuous strain can cause fracture.
- **Onset:** Slow onset of pain, in case of fracture sudden pain exacerbation.
- **Clinic:** Pain on the ulnar and volar aspect of the wrist, especially during ulnar duction and loading of Dig. IV/V.
- **Diagnosis:** Clinically and CT/MRI. Positive hamate stress test (see Chap. 5).
- **Therapy:** Initially conservative therapy, surgical therapy only in cases of long-term therapy failure.
- **Result:** Mostly complete healing under conservative therapy, occasionally surgical procedure (e.g., osteosynthesis, drilling, Herbert screw osteosynthesis, or resection of the hamate hook) necessary.

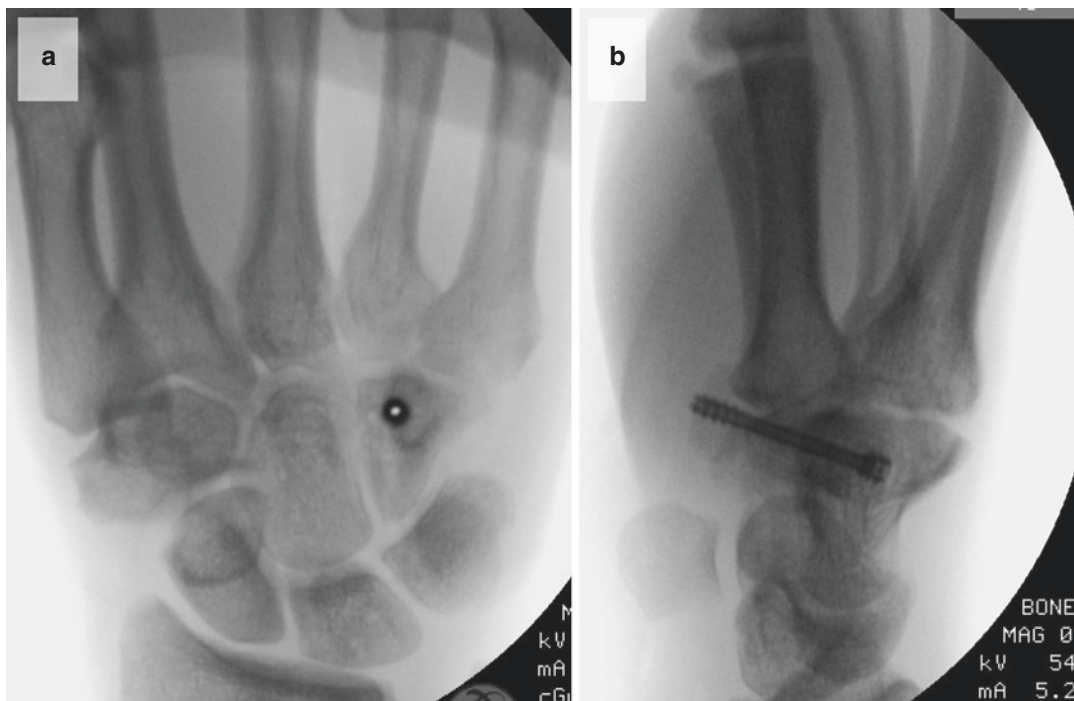


Fig. 7.5 Osteosynthesis of a nonunion of the hamate using a Herbert screw (**a**, a.p. projection; **b**, lateral projection)

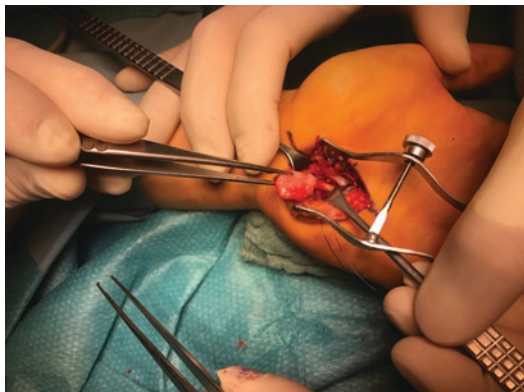


Fig. 7.6 Resection of the hamate hook following insufficient bony healing

7.5 Dupuytren's Contracture

A nodular remodeling reaction in the palm of the hand can be diagnosed frequently in climbers, without a known direct connection between climbing and the clinical picture of Dupuytren's contracture (see Fig. 7.7) [34–36]. However, it is assumed that recurrent microtrauma in the pal-

mar aponeurosis area caused by climbing could cause or accelerate corresponding remodeling reactions [1–3, 36].

In early stages, the disease manifests itself as a nodular hardening in the palm (usually Dig. IV/V); in advanced stages, it leads to retraction and limited extension of the affected fingers [34, 35].

Although individual cases have been described in young climbers (see Fig. 7.7), this disease is usually found in older athletes. Current figures show that in older climbing athletes (>35 years), this change in the palm is one of the ten most frequently diagnosed symptoms [4]. It is not known whether or to what extent climbing contributes to the genesis and the progress of the alteration reactions. In our own patient collective, the proportion of Dupuytren's contracture symptoms in athletes >35 years is 3%. The diagnosis (clinical diagnosis) and therapy of Dupuytren's contracture in climbers does not differ significantly from that in non-climbers [23, 34, 35].

Fortunately, the contracture is usually relatively painless for climbers and hinders only very few climbers. Additionally, the affected athletes

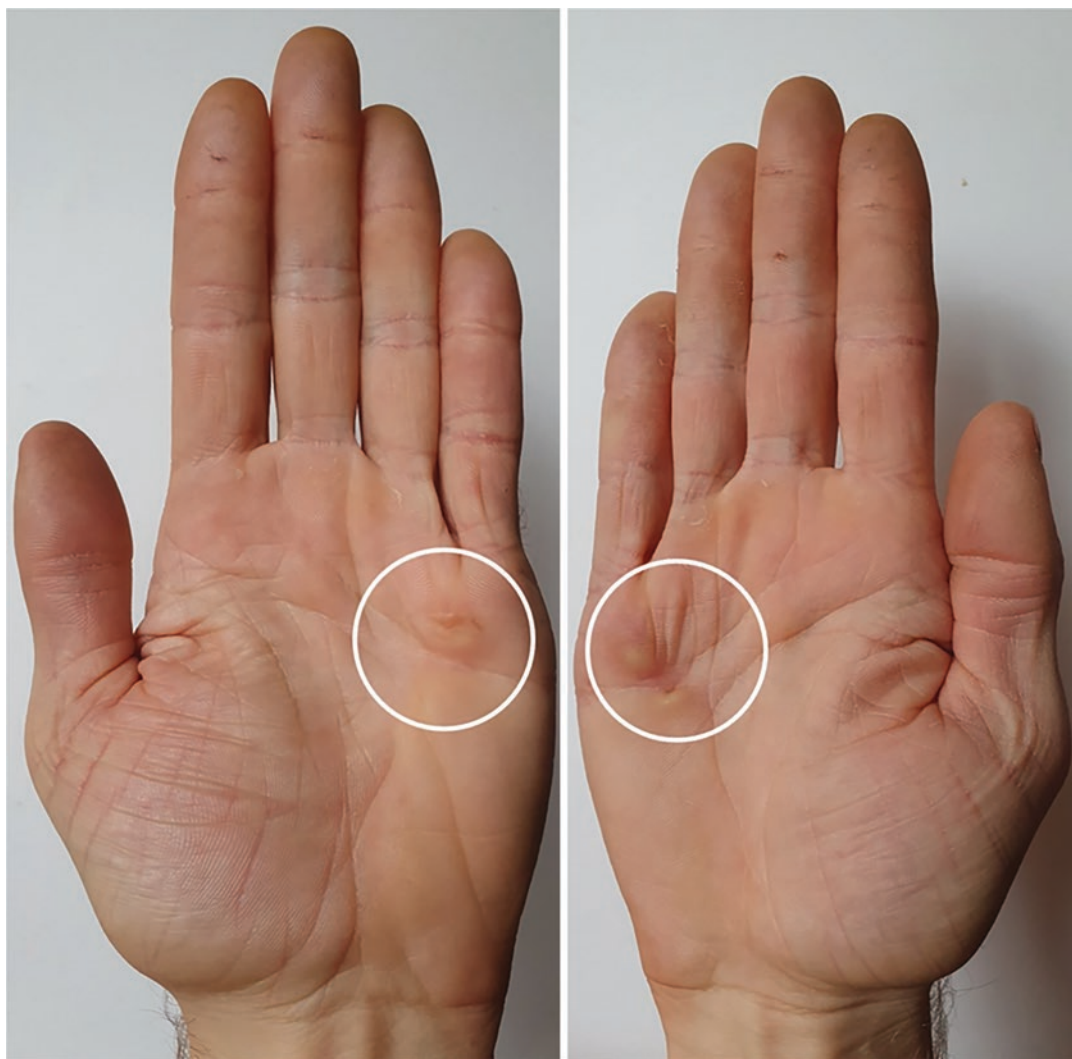


Fig. 7.7 A 34-year-old climber (former IFSC Boulder World cup athlete) who had developed an early stage Dupuytren's contracture on both hands over the last 10 years (Dig. IV/V)

rarely progress beyond the initial stage of the disease. In this initial stage, a “wait-and-see” or conservative approach can usually be taken. In case of (pressure) pain or disruptive movement restrictions, the surgical solution and resection of the scar strands is performed [34, 35].

- **Etiology:** Multifactorial, possible etiology due to climbing-related micro-trauma of the palmar aponeurosis

- **Onset:** Creeping hardening in the palm area
- **Clinic:** Nodular hardening of the palm (mainly Dig. IV/V)
- **Diagnosis:** Clinical
- **Therapy:** Conservative, in case of progress surgical resection of the scar strands
- **Result:** Initially often asymptomatic, in advanced stages limited climbing ability

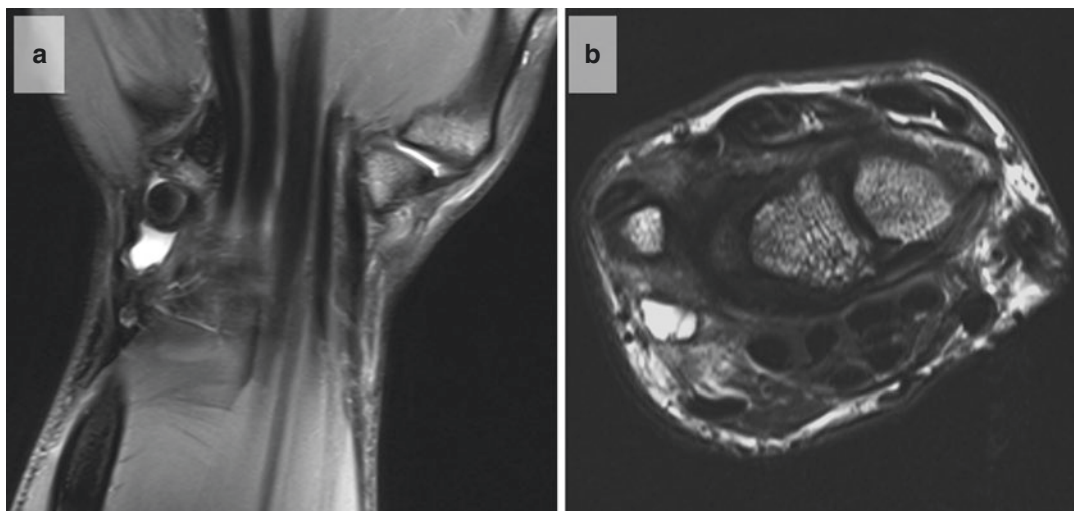


Fig. 7.8 MRI of an effusion with synovialitis in the distal radioulnar joint (**a**, coronary MRI image; **b**, axial projection)

7.6 Ulnocarpal Pain Syndromes

Ulnocarpal pain syndromes in climbers continue to be a challenge for treating physicians. These complaints can have various causes, all of which are difficult to diagnose. Additionally, the causes are difficult to distinguish from each other and are sometimes extremely difficult to treat. This often results in a lengthy diagnostic process and an often slow and frustrating course of therapy for affected athletes.

The following causes are discussed in connection with ulnar-sided wrist pain in climbers:

1. Lesions of the triangular fibrocartilaginous complex (TFCC lesions)
2. Instabilities or subluxations of the distal radioulnar joint (DRUJ)
3. Lesions of the lunotriquetral ligament [37]
4. Carpal instabilities (midcarpal instability) [38]
5. Triquetral impingement ligament tear (TILT) syndrome [39]
6. Overloading reactions around the extensor carpi ulnaris tendon
7. Synovitis with plica syndrome

For correct diagnosis, the physical examination is usually more precise than magnetic resonance imaging. The latter usually shows synovitis as a sign of stress reaction (Fig. 7.8). Clear structural causes for this are less often detectable. However, the reproducibility of the forces and loads occurring during climbing is clinically difficult and often, the patient's complaints can only be determined by taking a precise anamnesis. Reproduction of the pain symptoms is sometimes not possible. Nevertheless, a comparative examination of the opposite side should always be performed.

Climbers often describe a painful feeling of instability in the distal radioulnar joint (DRUJ). During stress in climbing, the force is usually transmitted via the carpus to the radius and ulna. The ligamentous apparatus of the TFCC forms one of the main stabilizers of the DRUJ. A corresponding load can therefore lead to synovitis in this area. The DRUJ also has little muscular stabilization, which favors the high load on the DRUJ. In the event of instability or subluxation of the DRUJ, overloading of the extensor carpi ulnaris tendon may also occur. Anatomical variations (especially ulna-plus/ulna-minus variations) can

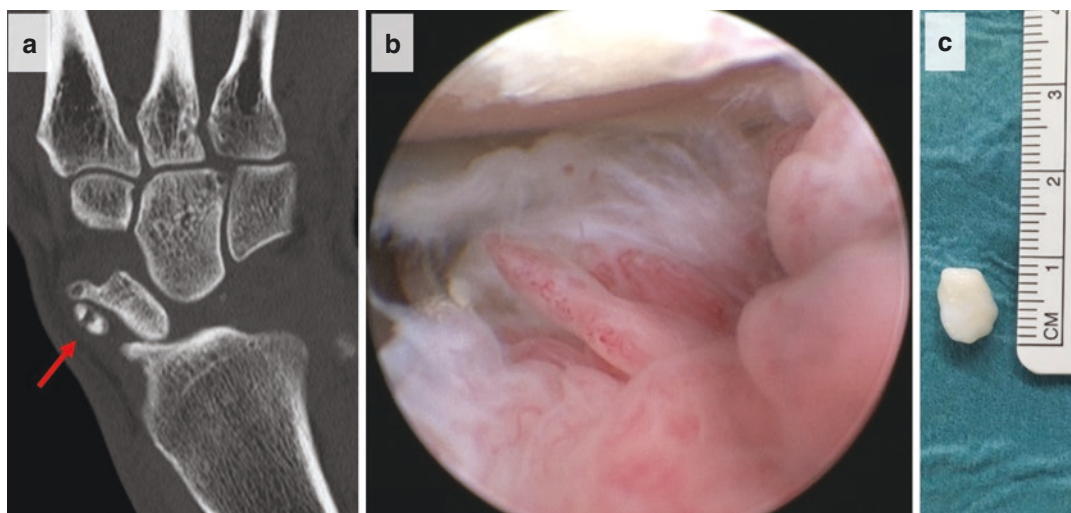


Fig. 7.9 A 43-year-old male climbing athlete with ulnar-sided wrist pain. CT scan shows an intra-articular body in the wrist joint (**a**, arrow). Arthroscopy of the wrist shows

synovialitis (**b**) and confirms the loose body within the joint (**c**, after resection)

promote the symptoms. Therapeutically, an initial conservative approach is usually recommended. The aim should be to reduce the load by adapting the training and avoiding the triggering hand positions. In case of persistent pain reactions and especially in case of clear synovitis confirmed by MRI, corticosteroid injections may be considered. Circular wrist tape often provides a feeling of stability for affected athletes. If therapy continues to fail and clinical/MRI suspicion of soft-tissue instability exists, wrist arthroscopy with synovectomy and, if necessary, stabilization of injured structures can be performed (Fig. 7.9).

- **Etiology:** Largely unclear. Suspected—TFCC lesions, DRUJ subluxations, lesions of the lunotriquetral ligament stump, midcarpal instabilities, “triangular impingement ligament tear (TILT)” syndrome and overloading reactions around the extensor carpi ulnaris tendon
- **Onset:** Usually a slow start

- **Clinic:** Stress-dependent ulnocarpal pain
- **Diagnosis:** Clinical, MRT
- **Therapy:** Conservative (taping), if necessary, injection therapy. Arthroscopy in case of therapy failure
- **Result:** Often moderate response to therapy

7.7 Conclusion

In summary, complaints around the metacarpus, wrist, and distal forearm form a large group of clinical pictures in climbers. With the appropriate level of knowledge about the underlying injuries and overuse damage, most complaints can be quickly and correctly diagnosed. Except for the ulnocarpal pain syndromes, most forms of injury respond well to the appropriate forms of therapy; however, healing processes are generally protracted, and stress reactions and adjustment of training after resumption of sport are unavoidable.

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